

Deep Patel

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PERSONAL PROFILE

Computer Science graduate, currently pursuing a Master's degree in Artificial Intelligence at BTU Cottbus (4th Semester). Hands-on experience writing clean Python code for data processing, text analysis, and standard machine learning algorithms. Fast learner comfortable working with standard IDEs, Git workflows, and data libraries. Seeking a student-level position, internship, or a 6-month master's thesis topic to gain practical industry experience.

EDUCATION

BTU Cottbus—Senftenberg

M.Sc. Artificial Intelligence (4th Semester)

2024 – Present

Cottbus, Germany

- **Current Academic Focus:** Machine Learning foundations, Data Science workflows, and NLP methods.
- **Objective:** Open to a 6-month industry thesis collaboration or practical student worker placements.

Parul University

B.Tech. Computer Science (Core Software Engineering)

2019 – 2023

Gujarat, India

- **Academic Foundations:** Object-Oriented Programming, Basic Data Structures, and SQL Databases.

ACADEMIC & DATA COMPUTING PROJECTS

Python ReAct Automation Agent Framework | *Groq API, Python*

LLM Logic Loop

- Programmed a library-free implementation of the Reasoning and Acting (ReAct) pattern using Python.
- Built and registered simple Python helper tools to calculate basic math expressions and count whitespace-separated words.
- Used regular expressions to parse the model output into actionable tool triggers and capture final text answers.

AI Resume Analyzer Pro | *Python, Streamlit, LLM API*

Generative AI Web Application

- Developed an interactive web application in Python using Streamlit to accept and parse text inputs from uploaded files.
- Integrated an LLM API client to evaluate user skill sets against specific job descriptions for basic gap analysis.
- Structured the UI cleanly with standard Streamlit widgets to display parsed document sections and generated text summaries side-by-side.

Fake News Classification Pipeline | *NLTK, Logistic Regression, Scikit-Learn*

Natural Language Processing

- Cleaned a textual dataset using Python's regular expressions library to remove special characters and lowercase text.
- Used the NLTK library to remove standard English stop-words and applied a PorterStemmer to reduce words to their roots.
- Converted the processed text into numeric features using TfidfTransformer and trained a standard Logistic Regression model.

California House Price Prediction Model | *XGBoost Regressor, Seaborn*

Data Regression

- Imported the California housing dataset via Scikit-Learn to build a basic price forecasting model.
- Generated a Seaborn correlation heatmap to visualize the linear relationships between the dataset's features.
- Split the data using an 80/20 train-test ratio and evaluated predictions using R2 score and Mean Absolute Error.

Automated Loan Status Prediction Engine | *Support Vector Machines (SVM), Pandas*

Predictive Modeling

- Processed a loan application dataset by dropping missing values and clean-encoding categorical strings into binary labels.
- Replaced custom alphanumeric symbols within the dataset columns to ensure smooth numerical formatting for training.
- Split the processed data with a stratified configuration and trained a Scikit-Learn linear SVM classifier.

PIMA Diabetes Diagnostics Classifier | *StandardScaler, SVM Card Matrix*

Binary Classification

- Analyzed physiological data attributes to classify records as either diabetic or non-diabetic.
- Standardized highly varying numerical features across the dataset rows using Scikit-Learn's StandardScaler tool.
- Partitioned data records into clean training and testing sets to fit a linear Support Vector Machine model.

SONAR Acoustic Reflectivity Discrimination Framework | *Logistic Regression, NumPy* Signal Processing Simulation

- Processed 60 discrete columns of numerical acoustic reflectivity data to identify objects as rocks or metallic mines.
- Grouped data samples by their class labels to compute feature averages and assess structural differences.
- Separated features and target labels into a 90/10 train-test split to evaluate a baseline Logistic Regression setup.

TECHNICAL SKILLS

Languages Syntaxes : Python, SQL, HTML5, CSS3, JavaScript

Libraries & Frameworks : Scikit-Learn, XGBoost, NumPy, Pandas, Natural Language Toolkit (NLTK), Seaborn, Matplotlib

Software Engineering Tools : Git, GitLab, Visual Studio Code (VS Code), PyCharm, Overleaf (LaTeX), Streamlit

Environments & Platforms : Google Colab, Jupyter Notebooks, MS Office Suite, Discord

Core Methodologies : Train-Test Splits, Baseline Data Preprocessing, Text Tokenization & Stemming, Tf-Idf Matrix Vectorization, Core Machine Learning Lifecycle

LANGUAGES

English : Fluent / Fließend (**IELTS Academic Score: 7.0** / CEFR C1 Equivalent)

German : A2/B1 (Active learner expanding vocabulary for quick integration into local teams)

Hindi / Gujarati : Native proficiency